ANSWER KEY & MARKING SCHEME SET 1

Chapter 7: **Particulate Nature of Matter** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
2	**(d)** spaces between particles	Particles of one fill the gaps.
3	**(b)** temperature increases the motion of particles	Faster particles.
4	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
5	**(b)** change of state	Latent heat.
6	**(d)** equal at equilibrium	Dynamic equilibrium.
7	**(c)** A is true but R is false.	A is true; R is false: heat is used to overcome forces between particles.
8	**(d)** Particles have the same kinetic energy but different arrangement	Same temperature, different state.
9	**(a)** Both A and R are true and R is the correct explanation of A.	Sugar particles occupy the inter-particle spaces of water.
10	**(a)** Both A and R are true and R is the correct explanation of A.	Faster particle motion increases diffusion.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Gas particles move rapidly in all directions (1); They spread uniformly and fill the container (1).	1+1
12	Evaporation is slow (1); Little cooling effect (1).	1+1
13	Only particles at the surface with enough energy escape (1); It occurs only at the liquid surface (1).	1+1
14	Reading stays at 0 degrees Celsius (1); Heat is used to change the state, not to raise temperature (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	-10 to 0: ice warms (1); 0 to 100: melting at 0 then water warms (1); At 100: boiling then steam warms to 110 (1).	1+1+1
16	Ice: least energy, close and fixed (1); Water: medium (1); Steam: highest, far apart (1).	1+1+1
17	It rises slightly (1); Dissolved particles hinder escape of water particles (1); More heat needed (1).	1+1+1
	OR: Hot water particles move faster (1); Higher average kinetic energy (1); Cold water particles move slower (1).	1+1+1
18	Number of particles remains the same (1); Only arrangement changes (1); Mass is the quantity of matter (1).	1+1+1
	OR: Particles in cold objects still move (1); They have kinetic energy (1); Heat flows from hot to cold objects (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Diffusion (1). (ii) The beaker at 65 degrees C (1). (iii) Particles of matter are always in motion and there are spaces between them (1). (iv) At a higher temperature particles have more kinetic energy and move faster, so they mix more quickly (2).	1+1+1+2
20	Liquid cools (1); Starts to solidify at the freezing point (1); Flat part: energy released as particles form a fixed structure (1); Temperature constant (1); Solid cools further (1).	1+1+1+1+1
	OR: Low temperature, high pressure favour solids and liquids (1); Dry ice sublimes at atmospheric pressure (1); Liquid CO_2 exists only at high pressure (1); Water changes with temperature at normal pressure (1); Boiling point depends on pressure (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that particles of a solid are at rest; they vibrate about fixed positions.
- Mixing up the spacing: gas particles are far apart, liquid and solid particles are close.
- Saying temperature keeps rising during melting or boiling; it stays constant during a change of state.

- Writing that evaporation happens only at the boiling point, or that it heats the surroundings; it causes cooling.
- Confusing diffusion (mixing of particles by themselves) with dissolving or with stirring.
- Stating that a liquid has a definite shape; it has a definite volume only.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 7: **Particulate Nature of Matter** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** water evaporates and takes heat from the air	Evaporation.
2	**(a)** spread it open in warm, dry, windy place	Raise evaporation.
3	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
4	**(b)** dissolved air coming out	Dissolved gases.
5	**(a)** sublimes, wax melts then remains	Different behaviours.
6	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
7	**(d)** sublimation	Both sublime.
8	**(d)** particles close and in fixed positions	Order.
9	**(c)** A is true but R is false.	A is true; R is false: particles in a solid only vibrate about fixed positions.
10	**(d)** A is false but R is true.	A is false (they vibrate in place); R is true.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Heat gently in a dish with an inverted funnel (1); Camphor sublimes and is collected; salt remains (1).	1+1
12	Raise: increase pressure (1); Lower: decrease pressure (1).	1+1
13	Extra heat is used for vaporisation (1); Boiling point stays constant at a given pressure (1).	1+1
14	Attractive forces are strong enough (1); Particles stay close but move around (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Vapour in air condenses on the cold glass (1); More vapour is present on humid days (1); Water does not come from inside the glass (1).	1+1+1
16	Liquid perfume evaporates (1); Vapour particles diffuse through the air (1); We smell them (1).	1+1+1
17	Large gaps in gases (1); Very small gaps in liquids (1); Particles in liquids resist being pushed closer (1).	1+1+1
	OR: Weaker forces between spirit particles (1); Particles escape easily (1); More heat is taken from the surface (1).	1+1+1
18	Different boiling points (1); Used in distillation (1); Vapour is condensed to collect (1).	1+1+1
	OR: More sublimation at higher temperature (1); Particles gain energy (1); Faster escape (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) 0 degrees C (1). (ii) 100 degrees C (1). (iii) 0 degrees C (1). (iv) The heat supplied is used to overcome the forces between particles and change the state from solid to liquid, not to raise the temperature (2).	1+1+1+2
20	Sun's heat evaporates water (1); Vapour rises, condenses to cloud (1); Rain returns water (1); Sweat evaporation cools us (1); Fast particles leave carrying energy (1).	1+1+1+1+1
	OR: Solids hold shape: tools, buildings (1); Liquids flow: water supply, oil (1); Gases compress: cylinders, balloons (1); Particle forces explain use (1); Choice of states in design (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that particles of a solid are at rest; they vibrate about fixed positions.
- Mixing up the spacing: gas particles are far apart, liquid and solid particles are close.
- Saying temperature keeps rising during melting or boiling; it stays constant during a change of state.
- Writing that evaporation happens only at the boiling point, or that it heats the surroundings; it causes cooling.
- Confusing diffusion (mixing of particles by themselves) with dissolving or with stirring.
- Stating that a liquid has a definite shape; it has a definite volume only.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 7: **Particulate Nature of Matter** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** particles of matter are stationary	Diffusion shows motion.
2	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
3	**(d)** gas particles are far apart	Gaps.
4	**(a)** vapour particles collide more with the walls	More collisions.
5	**(c)** A is true but R is false.	A is true; R is false: the faster particles leave.
6	**(c)** liquid or compressed gas expands and absorbs heat	Expansion cooling.
7	**(b)** heat is first used for changing state	Latent heat.
8	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
9	**(b)** Both A and R are true but R is not the correct explanation of A.	Both true, but the boiling point of water does not explain the cooling caused by evaporation.
10	**(a)** Both A and R are true and R is the correct explanation of A.	Large spaces between particles give a large volume.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	They gain kinetic energy (1); Temperature is a measure of average kinetic energy (1).	1+1
12	Plunger moves out (1); Heated gas expands as particles move faster (1).	1+1
13	Gas inside expands on heating (1); Particle speed increases and pressure increases (1).	1+1
14	Slow evaporation due to lower temperature (1); Humid air reduces it (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Temperature rises (1); More evaporation, vapour pressure increases (1); Condensation also occurs on cool walls (1).	1+1+1
16	Air particles gain energy (1); Pressure increases (1); Balloon stretches (1).	1+1+1
17	Warm air increases temperature (1); Moving air carries away vapour (1); Evaporation is faster (1).	1+1+1
	OR: Dry ice sublimes to CO2 gas and cools the air (1); Water vapour in air condenses (1); Tiny droplets appear as fog (1).	1+1+1
18	Particles are tightly packed (1); They only vibrate (1); Little movement through the solid (1).	1+1+1
	OR: Water vapour in air touches cold surfaces (1); It loses energy and forms ice crystals (1); Vapour to solid deposition (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Evaporation (1). (ii) Higher temperature, wind, larger surface area, lower humidity (any two) (1). (iii) Water seeps through the pores, evaporates from the outer surface and takes heat from the pot and water (1). (iv) A larger surface area lets more particles escape from the surface at the same time, speeding up evaporation (2).	1+1+1+2
20	Solid: fixed, vibrate (1); Melting: vibrations break pattern (1); Boiling: particles escape throughout (1); Condensation/freezing: loss of energy brings particles closer (1); Sublimation: direct escape from solid (1).	1+1+1+1+1
	OR: Dew (1); Cooling by sweat (1); Drying clothes (1); Camphor disappearing (1); Cloud formation (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that particles of a solid are at rest; they vibrate about fixed positions.
- Mixing up the spacing: gas particles are far apart, liquid and solid particles are close.
- Saying temperature keeps rising during melting or boiling; it stays constant during a change of state.

- Writing that evaporation happens only at the boiling point, or that it heats the surroundings; it causes cooling.
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